ANSIBLE – JENKINS DEPLOYMENT PROJECT

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Course: DevOps

Date: April 2026

ANSIBLE – JENKINS DEPLOYMENT PROJECT

Project Overview

This project demonstrates automated deployment using Jenkins integrated with Ansible.

The workflow includes:

- Developer pushes code to Git repository
- Jenkins pulls the code
- Jenkins builds the project and generates WAR file
- Jenkins triggers Ansible playbook
- Ansible deploys the application to production server
- Tomcat server runs the application

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Architecture / Workflow

1. Developer → Pushes code to Git
2. Jenkins → Pulls the code
3. Jenkins → Builds WAR file (myapp.war)
4. Jenkins → Triggers Ansible Playbook
5. Ansible → Connects to Production Node
6. Ansible → Copies WAR file to Tomcat
7. Ansible → Restarts Tomcat

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Tools Used

- Jenkins
- Ansible
- Apache Tomcat
- Maven
- Git
- AWS EC2

Execution Screenshots

- EC2 Instances (Ansible + Prod Node)
- Jenkins Pipeline Stages
- Ansible Playbook Execution
- Deployment Output
- Final Application Output (Browser screenshot)

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Challenges Faced

- SSH key configuration issues
- Ansible connectivity problems
- Jenkins integration setup
- Deployment errors

Conclusion

This project successfully demonstrates automated deployment using Jenkins and Ansible. It reduces manual effort and ensures faster and reliable application deployment.

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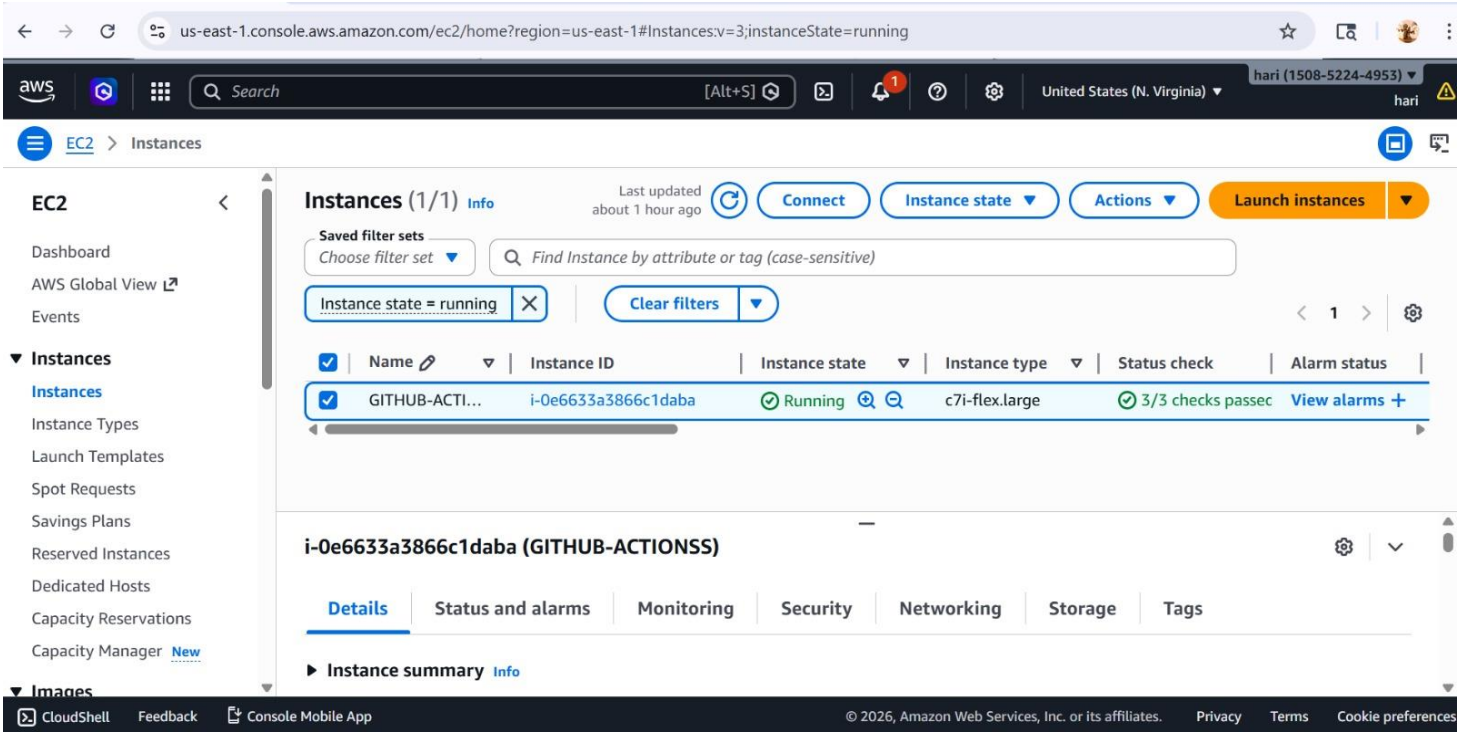
Important Note

The project was successfully executed on AWS EC2 instances.
Currently, the instances are terminated, so only screenshots are provided as proof.



```
\. "$HOME/.nvm/nvm.sh"

# Download and install Node.js:
nvm install 22
# Verify the Node.js version:
node -v # Should print "v22.18.0"
nvm current # Should print "v22.18.0"
# Verify npm version:
npm -v # Should print "10.9.3"
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 16631  100 16631    0     0   609k      0 --:--:-- --:--:-- --:--:--  624k
=> nvm is already installed in /home/ubuntu/.nvm, trying to update using git
=> => Compressing and cleaning up git repository

=> nvm source string already in /home/ubuntu/.bashrc
=> bash_completion source string already in /home/ubuntu/.bashrc
=> Close and reopen your terminal to start using nvm or run the following to use it now:

export NVM_DIR="$HOME/.nvm"
[ -s "$NVM_DIR/nvm.sh" ] && \. "$NVM_DIR/nvm.sh"  # This loads nvm
[ -s "$NVM_DIR/bash_completion" ] && \. "$NVM_DIR/bash_completion"  # This loads nvm bash_completion
Downloading and installing node v22.22.2...
Downloading https://nodejs.org/dist/v22.22.2/node-v22.22.2-linux-x64.tar.xz...
################################################################################################################################# 100.0%
Computing checksum with sha256sum
Checksums matched!
Now using node v22.22.2 (npm v10.9.7)
Creating default alias: default -> 22 (-> v22.22.2)
v22.22.2
v22.22.2
10.9.7
ubuntu@ip-172-31-88-39:~$
```

```
ubuntu@ip-172-31-88-39:~$ ./config.sh --url https://github.com/harinarayanag123-cmyk/java --token BWLUPDQQPOUZWNNYXJVRVS3J5HN74

--------------------------------------------------------------------------------
|        ____ _ _   _ _ _|_   _ |__     / \   ___| |_(_) ___  _ __  ___       |
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|                                                                              |
|                        Self-hosted runner registration                       |
|                                                                              |
--------------------------------------------------------------------------------

# Authentication


√ Connected to GitHub

# Runner Registration

Enter the name of the runner group to add this runner to: [press Enter for Default]

Enter the name of runner: [press Enter for ip-172-31-88-39]

This runner will have the following labels: 'self-hosted', 'Linux', 'X64'
Enter any additional labels (ex. label-1,label-2): [press Enter to skip]

√ Runner successfully added

# Runner settings

Enter name of work folder: [press Enter for _work]
```

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Not secure 52.90.178.23:9000/dashboard?id=amazon&selectedTutorial=github-actions

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Overview | Issues | Security Hotspots | Measures | Code | Activity | Project Settings | Project Information

1 Create GitHub Secrets

In your GitHub repository, go to **Settings > Secrets** and create two new secrets:

1. Click on **New repository secret**.
2. In the **Name** field, enter `SONAR_TOKEN`
3. In the **Value** field, enter an existing token, or a newly generated one: Generate a token
4. Click on **Add secret**.

1. Click on **New repository secret**.
2. In the **Name** field, enter `SONAR_HOST_URL`
3. In the **Value** field, enter `http://52.90.178.23:9000`
4. Click on **Add secret**.

Continue

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https://hub.docker.com/repositories/harinarayanag

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harinarayanag
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- Usage
  - Pulls
- Upgrade plan

Repositories

All repositories within the **harinarayanag** namespace.

Search by repository name | All content | Create a repository

Name	Last Pushed	Contains	Visibility	Scout
harinarayanag/myapp	2 days ago	IMAGE	Public	Inactive

1–1 of 1

https://nodejs.org/en/download

node

Download Node.js®

Get Node.js® v22.22.2 LTS for Linux using nvm with npm

Info Want new features sooner? Get the latest Node.js version instead and try the latest improvements!

```bash
# Download and install nvm:
curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.40.4/install.sh | bash

# in lieu of restarting the shell
\. "$HOME/.nvm/nvm.sh"

# Download and install Node.js:
nvm install 22

# Verify the Node.js version:
node -v # Should print "v22.22.2".

# Verify npm version:
npm -v # Should print "10.9.7".
```

Bash | Copy to clipboard

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https://github.com/harinarayanag123-cmyk/java/settings/actions/runners/new?arch=x64&os=linux

- Tags
- Rules
- Actions
  - General
  - Runners
  - OIDC
- Models (Preview)
- Webhooks
- Copilot
- Environments
- Codespaces
- Pages

Security and quality

- Advanced Security
- Deploy keys
- Secrets and variables

Integrations

already have an existing volume licensing agreement for your GitHub purchases, by downloading and configuring the GitHub Actions Runner, you agree to the GitHub Customer Agreement.

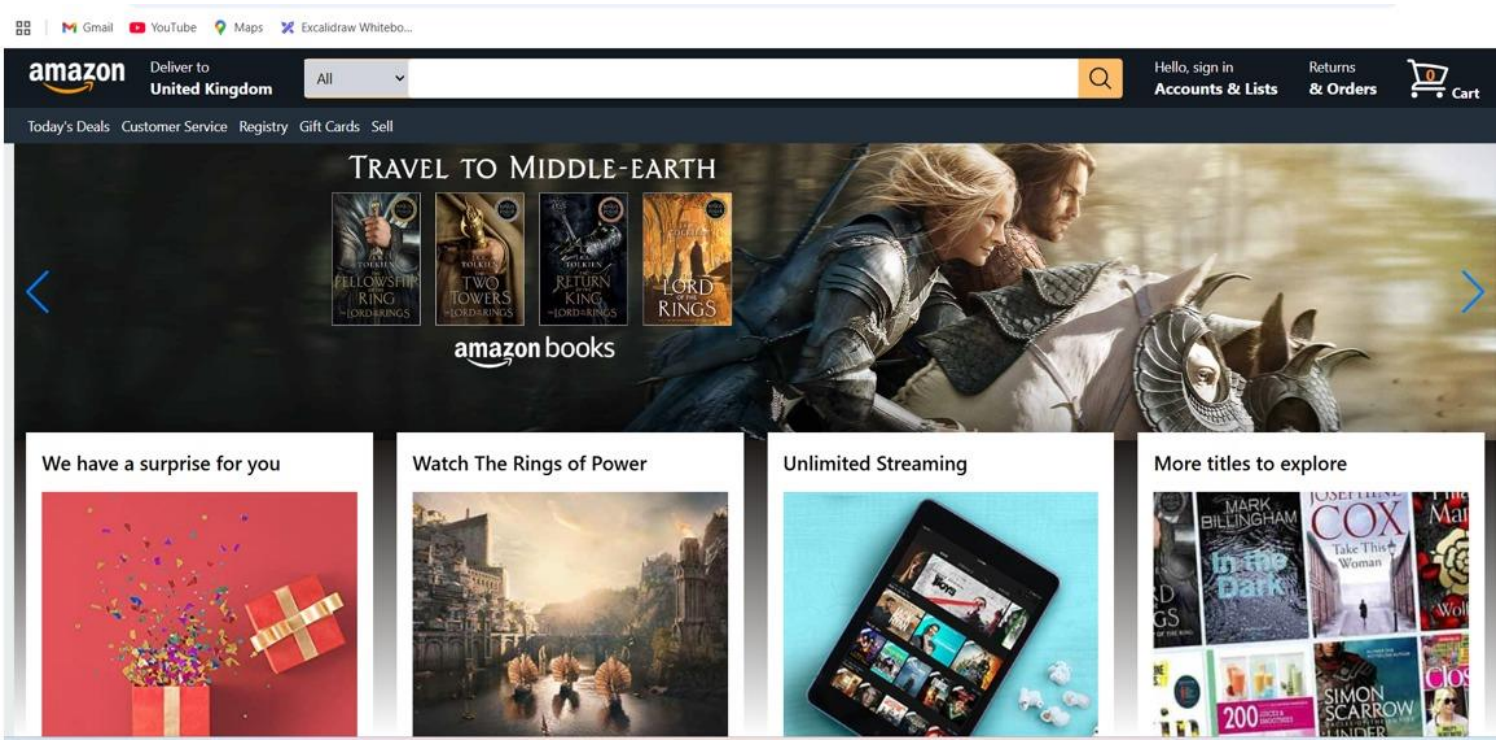
Runner image

macOS | Linux | Windows

Architecture

x64

Download

```
# Create a folder
$ mkdir actions-runner && cd actions-runner
# Download the latest runner package
$ curl -o actions-runner-linux-x64-2.333.1.tar.gz -L
https://github.com/actions/runner/releases/download/v2.333.1/actions-runner-linux-x64-2.333.1.tar.gz
# Optional: Validate the hash
$ echo "18f8f68ed1892854ff2ab1bab4fcaa2f5abeedc98093b6cb13638991725cab74  actions-runner-linux-x64-
2.333.1.tar.gz" | shasum -a 256 -c
# Extract the installer
$ tar xzf ./actions-runner-linux-x64-2.333.1.tar.gz
```

Configure